

Demo Contract

Group 8 - C.A.V.E: Cave Assessment and Visualization Equipment

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1 Contract Sections

The final project to be delivered and presented by Group 8 (C.A.V.E.) by April 2026 shall include the functionalities listed below:

1.1 Three Dimensional Mapping

The device shall produce a representation or projection of an enclosed three-dimensional space to a reasonable degree of accuracy, and should not rely on the presence of ambient or projected visible light.

1.2 Cost Effectiveness

The total cost of materials for the device shall stay within the project's \$750 limit and creating a minimally expensive device shall be a primary goal. Expensive 3-D LiDAR equipment will not be used.

1.3 Suitable for Caving

The device shall not restrict or hinder its operators movements unnecessarily, and should allow them to navigate small spaces. Use of the device should not fatigue the operator, and the system shall have sufficient endurance to map a large area.

1.4 Operator Safety

The device shall not create a significant risk of harm to the operator by, including but not limited to: interfering with safety equipment such as helmets and harnesses, dangerous exposed voltages, or improper battery storage.

1.5 Offline Operation

The device shall not require a connection to cellular or Wi-Fi networks in order to collect data from its environment. Internet connection is permissible and expected to upload raw data for processing into maps. The act of data collection will be 'tetherless', allowing the operator to roam freely.